Bioinformatics Internship Report

Task 1: DNA/Protein Sequence Analysis (BLAST Analysis)

1. Introduction

The primary objective of this task was to perform a sequence homology search using the Basic Local Alignment Search Tool (BLAST). By comparing a query protein sequence against a non-redundant database, we aim to identify homologous sequences, conserved domains, and evolutionary relationships.

2. Methodology

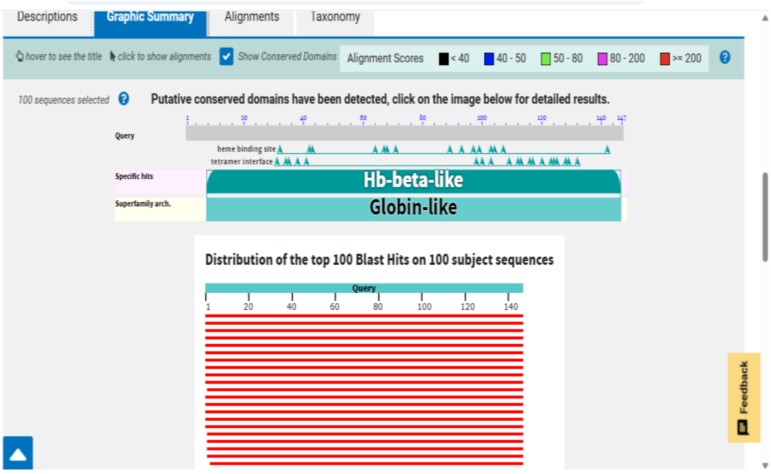
Program Used	BLASTp (Protein-Protein BLAST)
Database	Non-redundant protein sequences (nr)
Query ID	lcl|Query_3487745
Molecule Type	Amino Acid

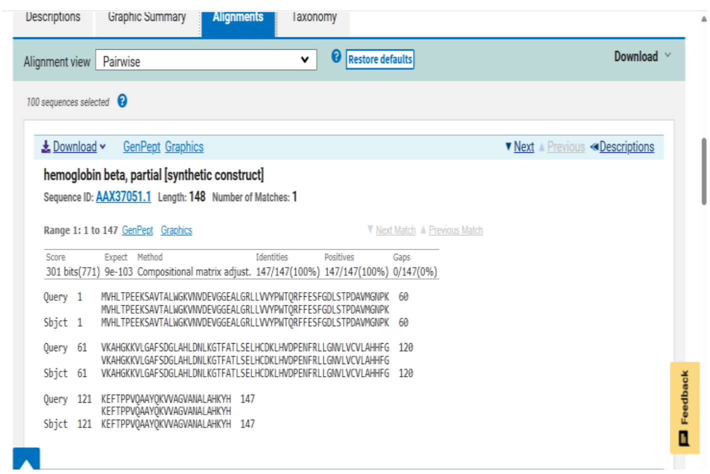
3. BLAST Results Analysis

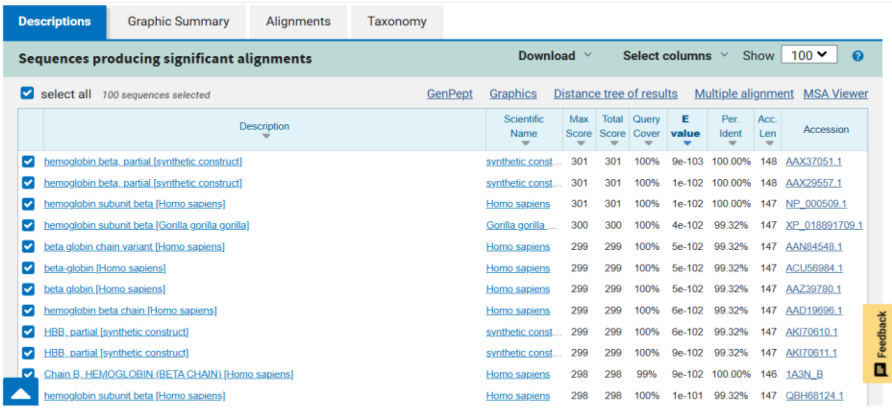
3.1. Graphic Summary & Conserved Domains

The graphic summary reveals that the query sequence belongs to the **Globin-like superfamily**. Specifically, a high-confidence match for the **Hb-beta-like** domain was detected. Key functional features identified include:

- **Heme binding sites:** Multiple conserved residues responsible for binding the prosthetic heme group.
- **Tetramer interface:** Residues involved in the interaction between different subunits of the hemoglobin molecule.







3.2. Significant Alignments (Top Hits)

The distribution of the top 100 hits shows a uniform red color across the entire query length (Score ≥ 200), indicating extremely high homology. The top sequences producing significant alignments include:

- **Hemoglobin subunit beta [Homo sapiens]:** 100% Query Coverage, 100% Identity, E-value: 1e-102.
- **Hemoglobin subunit beta [Gorilla gorilla gorilla]:** 100% Query Coverage, 99.32% Identity, E-value: 4e-102.

4. Results Interpretation

Statistical Significance: The E-values (ranging from 9e-103 to 1e-101) are exceptionally low, indicating that these matches are biologically significant and not due to random chance.

Sequence Identity: A 100% identity match with Human Hemoglobin Beta confirms the query sequence's origin. The 99.32% identity with Gorilla Hemoglobin further validates the highly conserved nature of this protein across primates.

5. Conclusion

The BLASTp analysis successfully identified the query sequence as Hemoglobin Subunit Beta. The high degree of conservation in functional domains (heme binding and tetramer interface) underscores its critical role in oxygen transport. This data serves as a foundation for subsequent tasks, including Multiple Sequence Alignment and 3D Structure Prediction.
